Equivalence of NFA with ε to NFA without ε

If L is accepted by NFA with ε transitions, then there exist L which is accepted by NFA without ε transitions.

Equivalence of NFA with ε to NFA without ε

In this method we try to remove all the ε transitions from given NFA. The method will be

1. Find out all the ε transitions from each state from Q. That will be called as ε-closure {q} where $q_i \in Q$.

2. Then δ' transitions can be obtained. The δ' transitions means an ε-closure on δ moves.

3. Step-2 is repeated for each input symbol and for each state of given NFA.

4. Using the resultant states the transition table for equivalent NFA without & can be built.

Theorem: If L is accepted by NFA with ε transitions, then there exist L which is accepted by NFA without ε transitions. AU May-09, Marks 12; Dec.-09, 13, Marks 8

Proof :

Let, $M = (Q, \Sigma, \delta, q_0, F)$ be an NFA with ε transitions.

Construct $M' = (Q, \Sigma, \delta', q_0, F')$ where

$F' = F \cup \{q_0\}$ if ε-closure contains state off

F otherwise

M' is a NFA without ε moves. The δ' function can be denoted by δ" with some input. For example, δ' (q, a) = δ" (q, a) for some q in Q and a from Σ. We will apply the method of induction with input x. The x will not be ε because

$\delta'(q_0, \varepsilon) = \{q_0\}$

8" (q_0, ε) = ε-closure (q_0). Therefore we will assume length of string to be 1.

Basis: |x| = 1. Then x is a symbol a.

$\delta'(q_0,a) = \delta''(q_0,a)$

Induction |x| > 1 Let x = wa

8'(q_0, w) = δ'(δ'(q_0, w), a)

By inductive hypothesis,

$\delta'(q_0, w) = \delta''(q_0, w) = p$

Now we will show that δ' (p, a) = δ (q_0, wa)

But

δ'(p, a) = U δ'(q, a) = U δ" (q, a)

q in p q in p

As

p = δ" (q_0, w)

We have, U δ" (q, a) = δ" (q_0, wa)

q in pExample for Understanding

Example 1.9.1 *Convert the given NFA with ε to NFA without ε.*

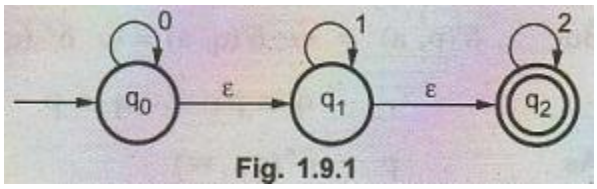


Fig. 1.9.1

Solution: Step 1: We will first obtain ε - closure of each state i.e. we will find out ε - reachable states from current state.

Hence

ε - closure (q_0) = {q_0, q_1, q_2}

ε - closure (q_1) = {q_1, q_2}

ε - closure (q_2) = {q_2}

As ε - closure (q_0) means with null input (no input symbol) we can reach to q_0, q_1 or q_2. In a similar manner for q_1 and q_2 ε - closures are obtained.

Step 2: Now we will obtain δ' transitions for each state on each input symbol.

δ' (q_0, 0) = ε - closure (δ (δ' (q_0, ε), 0))

Discover more

Poriyaan - Anna University

CSE Notes Subscription

Engineering Chemistry Notes

= ε - closure (δ(ε - closure(q_0), 0))

= ε - closure (δ(q_0, q_1, q_2), 0).

$= \varepsilon$ - closure ($\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2,0)$)

$= \varepsilon$ - closure ($q_0 \cup \phi \cup \phi$)

$= \varepsilon$ - closure (q_0) = $\{q_0, q_1, q_2\}$

$\delta'(q_0, 1) = \varepsilon$ - closure ($\delta(\delta`(q_0,\varepsilon), 1)$)

$= \varepsilon$ - closure ($\delta(q_0, q_1, q_2), 1$)

$= \varepsilon$ - closure ($\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2,1)$)

$= \varepsilon$ closure ($\phi \cup q_1 \cup \phi$)

$= \varepsilon$ - closure (q_1)

$\delta'(q_0, 1) = \{q_1, q_2\}$

$\delta'(q_1,0) = \varepsilon$ - closure ($\delta(\delta`(q_1,\varepsilon), 0)$)

$= \varepsilon$ - closure ($\delta(\varepsilon$ - closure (q_1), 0))

$= \varepsilon$ closure ($\delta(q_1,q_2), 0$)

$= \varepsilon$ closure ($\delta(q_1, 0) \cup \delta(q_2, 0)$)

$= \varepsilon$ closure ($\phi U \phi$)

$= \varepsilon$ - closure (ϕ)

$= \phi$

$\delta'(q_1, 1) = \varepsilon$ - closure ($\delta(\delta`(q_1, \varepsilon), 1)$)

$= \varepsilon$ - closure ($\delta(\varepsilon$ - closure (q_1), 1))

$= \varepsilon$ - closure (δ (q_1, q_2), 1)

$= \varepsilon$ - closure (δ (q_1, 1) U δ (q_2, 1))

$= \varepsilon$ - closure (q_1 U ϕ)

$= \varepsilon$ - closure (q_1)

$= \{q_1, q_2\}$

$\delta'(q_2, 0) = \varepsilon$- closure ($\delta$ (δ` (q_2, ε), 0))

$= \varepsilon$ - closure (δ (ε - closure (q_2), 0))

$= \varepsilon$ - closure (δ (q_2, 0))

$= \varepsilon$ - closure (ϕ)

$= \phi$

δ ' (q_2, 1) = ε - closure (δ (δ` (q_2, ε), 1))

$= \varepsilon$ closure (δ (ε - closure (q_2), 1))

$= \varepsilon$ - closure (δ (q_2,1))

$= \varepsilon$ - closure (ϕ)

δ' (q_2, 1) = ϕ

δ' (q_0, 2) = ε - closure (δ (δ` (q_2, ε), 2))

$= \varepsilon$ - closure (δ (ε -closure (q_0), 2))

$= \varepsilon$ - closure (δ (q_0, q_1, q_2), 2).

$= \varepsilon$ - closure (δ (q_0, 2) U δ (q_1, 2) U δ (q_2, 2))

$= \varepsilon$ - closure (q_2)

$= \varepsilon$ - closure (q_2)

$= \{q_2\}$

δ' (q_1,2) $= \varepsilon$ - closure (δ (δ` (q_1, ε), 2))

$= \varepsilon$ - closure (δ (ε - closure (q_1), 2))

$= \varepsilon$ - closure (δ (q_1, q_2), 2)

= &- closure (δ (q_1, 2) U δ (q_2, 2))

$= \varepsilon$ closure (ϕ U q_2)

$= \{q_2\}$

δ' (q_2, 2) $= \varepsilon$ - closure (δ (δ` (q_2, ε), 2))

$= \varepsilon$ - closure (δ (ε - closure (q_2), 2))

$= \varepsilon$ - closure (δ (q_2, 2))

$= \varepsilon$ - closure (q_2)

$= \{q_2\}$

Now we will summarize all the computed δ' transitions -

$\delta'(q_0, 0) = \{q_0, q_1, q_2\}$, $\delta'(q_0, 1) = \{q_1, q_2)$, $\delta'(q_0, q_2) = \{q_2\}$

$\delta'(q_1, 0) = \phi$ $\delta'(q_1, 1) = \{q_1, q_2\}$,$\delta'(q_1, 2) = \{q_2\}$

$\delta'(q_2, 0) = \phi$ $\delta'(q_2, 1) = \phi$ $\delta'(q_2, 2) = \{q_2\}$

Step 3:

From this we can write the transition table as -

State \ Input	0	1	2
q_0	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
q_1	ϕ	$\{q_1, q_2\}$	$\{q_2\}$
q_2	ϕ	ϕ	$\{q_2\}$

Step 4:

The NFA will be,

Here q_0, q_1 and q_2 is a final state because ε closure (q_0), ε - closure (q_1) and ε - closure (q_2) contains final state q_2.

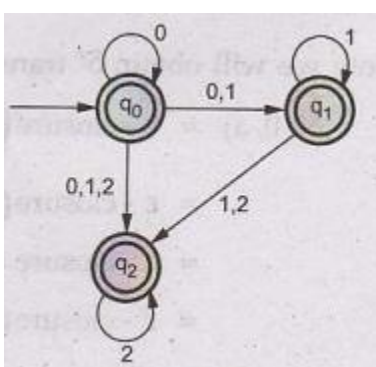


Fig. 1.9.2

Thus by definition δ"

$\delta'(q_0, wa) = \delta''(q_0, wa)$

Rule for conversion

$\delta'(q, a) = \varepsilon$ - closure $(\delta(\delta(q, \varepsilon), a))$

where $\delta(q,\varepsilon) = \varepsilon$ - closure (q)

Example 1.9.2 *Convert the following NFA with e to NFA without ε.*

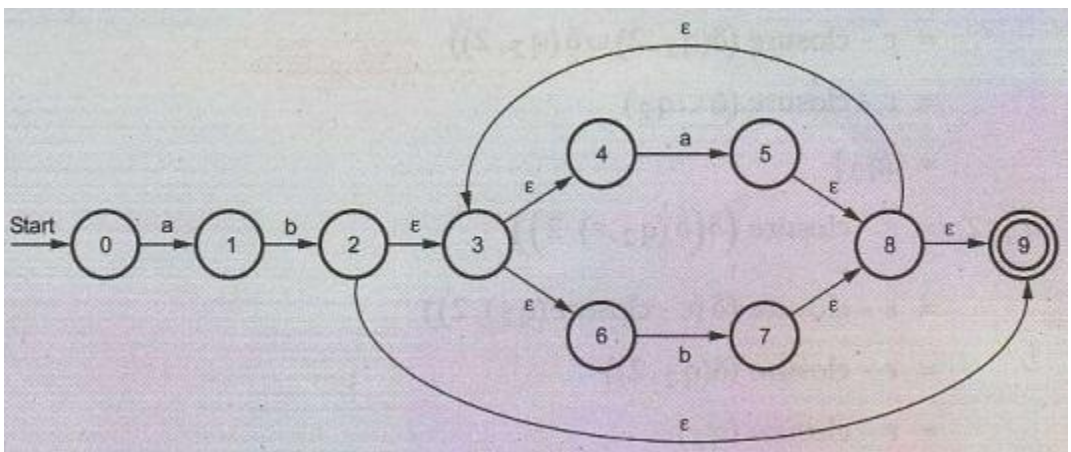


Fig. 1.9.3

Solution: We will first obtain ε - closure of every state. The ε - closure is basically an ε - transition from one state to other. Hence

ε - closure (0) = {0}

ε - closure (1) = {1}

ε - closure (2) = {2, 3, 4, 6, 9}

ε - closure (3) = {3, 4, 6}

ε - closure (4) = {4}

ε - closure (5) = {5, 8, 3, 4, 6, 9}

= {3, 4, 5, 6, 8, 9} sorted it!

ε - closure (6) = {6}

ε - closure (7) = {7, 8, 3, 4, 6, 9}

= {3, 4, 6, 7, 8, 9}

ε - closure (8) = {8, 3, 4, 6, 9}

= {3, 4, 6, 8, 9}

ε - closure (9) = {9}

Now we will obtain δ' transitions for each state and for each input symbol.

δ' (0, a) = ε - closure (δ (δ` (0, ε), a))

= ε - closure (δ (ε - closure(0), a))

= ε - closure (δ (0, a))

= ε - closure (1)

{1}

δ ' (0, b) = ε - closure (δ (δ` (0,ε), b))

$= \varepsilon$ - closure (δ (ε - closure (0), b))

$= \varepsilon$ - closure (δ (0, b))

$= \varepsilon$ - closure (ϕ)

$= \phi$

δ' (1,a) $= \varepsilon$ - closure (δ (δ` (1,ε), a))

$= \varepsilon$ - closure (δ (ε - closure (1), a))

$= \varepsilon$ - closure (δ (1, a))

$= \varepsilon$ - closure (ϕ)

$= \phi$

δ ' (1, b) $= \varepsilon$ - closure (8 (8 (1,ε), b))

$= \varepsilon$ - closure (δ (ε - closure (1), b))

$= \varepsilon$ - closure (δ (1, b))

$= \varepsilon$ - closure (2)

$= \{2, 3, 4, 6, 9\}$

δ ' (2, a) $= \varepsilon$ - closure (δ (δ` (2,ε), a))

$= \varepsilon$ - closure (δ (ε - closure (2), a))

$= \varepsilon$ - closure (δ (2, 3, 4, 6, 9), a)

$= \varepsilon$ - closure (δ (2, a) U δ (3, a) U δ (4, a) U δ (6, a) U δ (9, a))

$= \varepsilon$ - closure (ϕ U ϕ U 5 U ϕ U ϕ)

$= \varepsilon$ - closure (5)

= {3, 4, 5, 6, 8, 9}

δ' (2, b) = ε - closure: (δ (δ (2, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (2), b))

$= \varepsilon$ - closure (δ (2, 3, 4, 6, 9), b)

$= \varepsilon$ - closure (δ(2, b) U δ(3, b) U δ(4, b) U δ(6, b) U δ(9, b))

$= \varepsilon$ - closure (ϕ U ϕ U ϕ U 7 U ϕ)

$= \varepsilon$ - closure (7)

= {3, 4, 6, 7, 8, 9}

δ'(3, a) = ε - closure (δ (δ'(3,ε), a))

$= \varepsilon$ - closure (δ (ε - closure (3), a))

$= \varepsilon$ - closure (δ (3, 4, 6), a).

$= \varepsilon$ - closure (δ (3, a) U δ (4, a) U δ (6, a))

$= \varepsilon$ - closure (ϕ U 5 U ϕ)

$= \varepsilon$ - closure (5)

$= \{3, 4, 5, 6, 8, 9\}$

$\delta`(3, b) = \varepsilon$ - closure $(\delta (\delta`(3, \varepsilon), b))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(3), b))$

$= \varepsilon$ - closure $(\delta (3, 4, 6), b)$

$= \varepsilon$ - closure $(\delta (3, b) \cup \delta (4, b) \cup \delta (6, b))$

$= \varepsilon$ - closure $(\phi \cup \phi \cup 7)$

$= \varepsilon$ - closure (7)

$= \{3, 4, 6, 7, 8, 9\}$

$\delta` (4, a) = \varepsilon$ - closure $(\delta (\delta` (4, \varepsilon), a))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(4), a))$

$= \varepsilon$ - closure $(\delta (4, a))$

$= \varepsilon$ - closure (5)

$= \{3, 4, 5, 6, 8, 9\}$

$\delta`(4, b) = \varepsilon$ - closure $(\delta (\delta` (4, \varepsilon), b))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(4, b))$

$= \varepsilon$ - closure $(\delta (4, b))$

$= \varepsilon$ - closure (ϕ)

$= \phi$

$\delta'(5, a) = \varepsilon$ - closure (δ (δ`(5, ε), a))

$= \varepsilon$ - closure (δ (ε - closure (5), a))

$= \varepsilon$ - closure (δ (3, 4, 5, 6, 8, 9), a)

= e-closure (δ (3, a) U (4, a) U δ (5, a) U δ (6, a) U δ (8, a) U δ (9,a))

$= \varepsilon$ - closure (ϕ U 5 U ϕ U ϕ U ϕ U ϕ U ϕ)

$= \varepsilon$ - closure (5)

= {3, 4, 5, 6, 8, 9}

δ' (5, b) = ε - closure (δ (δ`(5, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (5), b))

$= \varepsilon$ - closure (δ (3, 4, 5, 6, 8, 9), b)

$= \varepsilon$ - closure (δ (3, b) U δ (4, b) U δ (5, b) U δ (6, b) U δ (8, b) U δ (9, b))

$= \varepsilon$ - closure (ϕ U ϕ U ϕ U 7 U ϕ U ϕ)

$= \varepsilon$ - closure (7)

= {3, 4, 6, 7, 8, 9}

δ' (6, a) = ε - closure (δ (δ` (6, ε), a))

$= \varepsilon$ - closure (δ (ε - closure (6), a))

$= \varepsilon$ - closure (δ (6, a))

$= \varepsilon$ - closure (ϕ)

$= \phi$

8' (6, b) $= \varepsilon$ - closure (δ (δ (6, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (6), b))

$= \varepsilon$ - closure (δ (6, b))

$= \varepsilon$ - closure (7)

= {3, 4, 6, 7, 8, 9}

δ' (7, a) $= \varepsilon$ - closure (δ (δ`(7, ε), a))

$= \varepsilon$ - closure (δ (ε - closure (7), a))

$= \varepsilon$ - closure (δ (3, 4, 6, 7, 8, 9), a)

$= \varepsilon$ - closure (δ (3, a) U δ (4, a) U δ (6, a) U δ (7, a) U δ (8, a) U δ (9, a))

$= \varepsilon$ - closure (ϕ U 5 U ϕ U ϕ U ϕ U ϕ)

$= \varepsilon$ - closure (5)

= {3, 4, 5, 6, 8, 9}

δ ' (7, b) $= \varepsilon$ - closure (δ (δ` (7, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (7), b))

$= \varepsilon$ - closure (δ (3, 4, 6, 7, 8, 9), b)

$= \varepsilon$ - closure (δ (3, b) U δ (4, b) U δ (6, b) U δ (7, b) U δ (8, b) U δ (9, b))

$= \varepsilon$ - closure (ϕ U ϕ U 7 U ϕ U ϕ U ϕ)

$= \varepsilon$ - closure (7)

= {3, 4, 6, 7, 8, 9}

δ' (8, a) = ε - closure (δ (δ`(8, ε), a))

$= \varepsilon$ - closure (δ (ε - closure (8), a))

$= \varepsilon$ - closure (δ (3, 4, 6, 8, 9), a)

$= \varepsilon$ - closure (δ (3, a) U δ (4, a) U δ (6, a) U δ (8, a) U δ (9, a))

$= \varepsilon$ - closure (ϕ U 5 U ϕ U ϕ U ϕ)

$= \varepsilon$ - closure (5)

= {3, 4, 5, 6, 8, 9}

δ' (8, b) = ε - closure (δ (δ` (8, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (8), b))

$= \varepsilon$ - closure (δ (3, 4, 6, 8, 9), b)

$= \varepsilon$ - closure (δ (3, b) U δ (4, b) U δ (6, b) U δ (8, b) U δ (9, b))

$= \varepsilon$ - closure (ϕ U ϕ U 7 U ϕ U ϕ)

$= \varepsilon$ - closure (7)

$= \{3, 4, 6, 7, 8, 9\}$

$\delta'(9, a) = \varepsilon$ - closure (δ (δ` (9, ε), a))

$= \varepsilon$ - closure (δ (ε - closure (9), a))

$= \varepsilon$ - closure (δ (9, a))

$= \varepsilon$ - closure (ϕ)

$= \phi$

$\delta'(9, b) = \varepsilon$ - closure (δ (δ` (9, ε), b))

$= \varepsilon$ - closure (δ (ε - closure (9), b))

$= \varepsilon$ - closure (δ (9, b))

$= \varepsilon$ - closure (ϕ)

$= \phi$

Now we will build the transition table using above calculated δ' transitions.

State \ Input	a	b
{0}	{1}	ϕ
{1}	ϕ	{2, 3, 4, 6, 9}
{2}	{3, 4, 5, 6, 8, 9}	{3, 4, 6, 7, 8, 9}
{3}	{3, 4, 5, 6, 8, 9}	{3, 4, 6, 7, 8, 9}
{4}	{3, 4, 5, 6, 8, 9}	ϕ
{5}	{3, 4, 5, 6, 8, 9}	{3, 4, 6, 7, 8, 9}
{6}	ϕ	{3, 4, 6, 7, 8, 9}
{7}	{3, 4, 5, 6, 8, 9}	{3, 4, 6, 7, 8, 9}
{8}	{3, 4, 5, 6, 8, 9}	{3, 4, 6, 7, 8, 9}
{9}	ϕ	ϕ

State {2} is a final state since ε - closure (2) contains 9 which is actually a final state in given NFA. Similarly state 5, 7, 8 and 9 are final states.

Example 1.9.3 Convert the following NFA with & to NFA without ε.

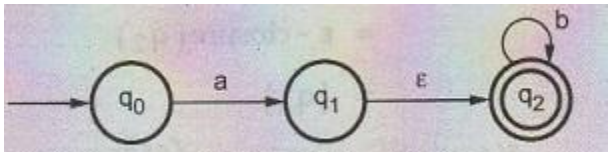


Fig. 1.9.4

Solution: We will first obtain & - closures of q_0, q_1 and q_2 as follows.

ε - closure (q_0) = {q_0}

ε - closure (q_1) = {q_1, q_2}

ε - closure (q_2) = $\{q_2\}$

Now the δ' transitions on each input symbol is obtained as,

$8'$ (q_0, a) = ε - closure (δ (δ` (q_0, ε), a))

= ε - closure (δ (ε - closure (q_0), a))

= ε - closure (δ (q_0, a))

= ε - closure (q_1)

= $\{q_1, q_2\}$

δ' (q_0, b) = ε - closure (δ (δ` (q_0, ε), b))

= ε - closure (δ (ε - closure (q_0), b))

= ε - closure (δ (q_0, b)).

= ϕ

δ' (q_1, a) = ε - closure (δ (δ` (q_1, ε), a))

= ε - closure (δ (ε - closure (q_1), a))

= ε - closure (δ (q_1, q_2), a)

= ε - closure (δ (q_1, a) U δ (q_2, a))

= ε - closure (ϕ U ϕ)

= ϕ

$\delta'(q_1, b) = \varepsilon - closure(\delta(\delta`(q_1, \varepsilon), b))$

$= \varepsilon - closure(\delta(\varepsilon - closure(q_1), b))$

$= \varepsilon - closure(\delta(q_1, q_2), b)$

$= \varepsilon - closure(\delta(q_1, b) \cup \delta(q_2, b))$

$= \varepsilon - closure(q_2)$

$= \varepsilon - closure(q_2)$

$= \{q_2\}$

$\delta'(q_2, a) = \varepsilon - closure(\delta(\delta`(q_2, \varepsilon), a))$

$= \varepsilon - closure(\delta(\varepsilon - closure(q_2), a))$

$= \varepsilon - closure(\delta(q_2, a))$

$= \varepsilon - closure(\phi)$

$= \phi$

$\delta'(q_2, b)$

$= \varepsilon - closure(\delta(\delta`(q_2, \varepsilon), b))$

$= \varepsilon - closure(\delta(\varepsilon - closure(q_2), b))$

$= \varepsilon - closure(\delta(q_2, b))$

$= \varepsilon - closure(q_2)$

= {q_2}

The transition table can be –

State \ Input	a	b
→ q_0	$\{q_1, q_2\}$	ϕ
q_1	ϕ	$\{q_2\}$
q_2	ϕ	$\{q_2\}$

States q_1 and q_2 becomes the final as closures of q_1 and q_2 contains the final state q_2. The NFA can be shown by transition diagram as shown in Fig. 1.9.5.

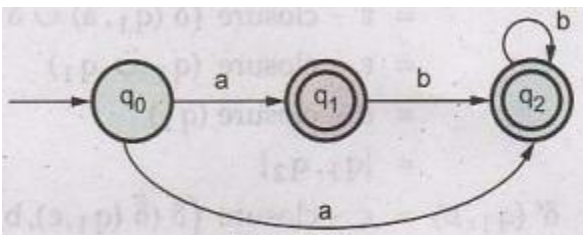


Fig. 1.9.5

Example 1.9.4 Convert the given NFA with ε to ordinary NFA.

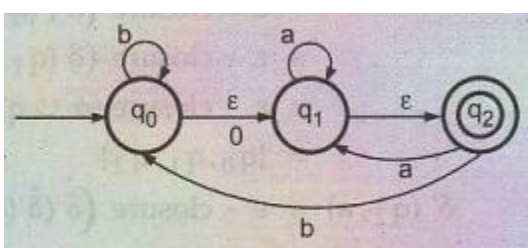


Fig. 1.9.6

Solution: We will first obtain ε - closure of each state.

ε - closure (q_0) = {q_0, q_1, q_2}

ε - closure (q_1) = {q_1, q_2)

ε - closure (q_2) = {q_2}

Now &' transitions for each state on each input.

8' (q_0, a) = ε - closure (δ (δ` (q_0, ε), a))

= ε - closure (δ (ε - closure (q_0), a))

= ε closure (δ ((q_0, q_1, q_2), a))

= ε closure (δ (q_0, a) U δ (q_1, a) U δ (q_2, a))

= ε - closure (ϕ U q_1 U q_1).

= ε - closure (q_1)

= (q_1, q_2)

δ' (q_0, b) = ε - closure (δ (δ` (q_0, ε), b))

= ε closure (δ (ε - closure (q_0), b))

= ε - closure (δ ((q_0, q_1, q_2), b))

= ε - closure (δ (q_0, b) U δ (q_1, b) U δ (q_2, b))

= ε - closure (q_0 U ϕ U q_0)

= ε - closure (q_0)

$= (q_0, q_1, q_2)$

$\delta' (q_1, a) = \varepsilon$ - closure $(\delta (\delta` (q_1, \varepsilon), a))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(q_1), a))$

$= \varepsilon$ - closure $(\delta (\{q_1, q_2), a))$

$= \varepsilon$ - closure $(\delta (q_1, a) \text{ U } \delta (q_2, a))$

$= \varepsilon$ - closure $(q_1 \text{ U } q_1)$

$= \varepsilon$ - closure (q_1)

$= \{q_1, q_2\}$

$\delta' (q_1, b) = \varepsilon$ - closure $(\delta (\delta` (q_1, \varepsilon), b))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(q_1), b))$

$= \varepsilon$ - closure $(\delta (q_1, q_2), b))$

$= \varepsilon$ - closure $(\delta (q_1, b) \text{ U } \delta (q_2, b))$

$= \varepsilon$ closure $(\phi \text{ U } q_0)$

$= \{q_0, q_1, q_2\}$

$\delta' (q_2, a) = \varepsilon$ - closure $(\delta (\delta` (q_2, \varepsilon), a))$

$= \varepsilon$ - closure $(\delta (\varepsilon$ - closure $(q_2), a))$

$= \varepsilon$ - closure $(\delta (q_2, a))$

= ε - closure (q_1)

= (q_1, q_2)

δ' (q_2, b) = ε - closure (δ (δ` (q_2, ε), b))

= ε closure (δ (ε - closure (q_2), b))

= ε - closure (δ (q_2, b))

= ε - closure (q_0)

= {q_0, q_1, q_2}

The transition table can be –

	a	b
→ q_0	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
q_2	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$

Example 1.9.5 Construct an NFA without ε transitions for the NFA given in Fig. 1.9.7. AU May-12, Marks 8

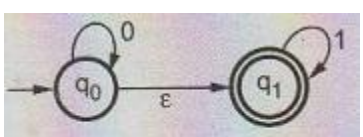


Fig. 1.9.7

Solution: We will first obtain ε - closures of q_0 and q_1 as follows -

ε - Closure (q_0) = $\{q_0, q_1\}$

ε - Closure (q_1) = $\{q_1\}$

Now δ' transitions on each input symbol is obtained as -

$\delta'(q_0, 0) = \varepsilon$ - Closure (δ (δ` (q_0, ε), 0))

= ε - Closure (δ (ε - Closure (q_0), 0))

= ε - Closure (δ ((q_0, q_1), 0))

= ε - Closure (q_0)

δ' (q_0, 0) = $\{q_0, q_1\}$

δ' (q_0, 1) = ε - Closure (δ (δ` (q_0, ε), 1))

= ε - Closure (δ (ε - Closure (q_0), 1))

= ε - Closure (8 ((q_0, q_1), 1))

= ε - Closure (q_1)

δ' (q_0, 1) = $\{q_1\}$

δ' (q_1, 0) = ε - Closure (δ (δ` (q_1, ε), 0))

= ε - Closure (δ (ε - Closure (q_1), 0))

= ε - Closure (δ (q_1, 0))

= ε - Closure (ϕ)

$\delta'(q_1, 0) = \phi$

$\delta'(q_1, 1) = \varepsilon$ - Closure (δ (δ` (q_1, ε), 1))

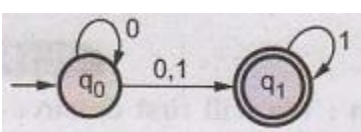
$= \varepsilon$ - Closure (δ (ε - Closure (q_1), 1))

$= \varepsilon$ - Closure (δ (q_1, 1))

$= \varepsilon$ - Closure (q_1)

$\delta'(q_1, 1) = \{q_1\}$

The transation table for NFA without & will be,

		0	1
→	q_0	$\{q_0, q_1\}$	$\{q_1\}$
*	q_1	ϕ	$\{q_1\}$

The transition diagram will be -